



الأكاديمية السعودية الرقمية
SAUDI DIGITAL ACADEMY



Big Data & AI Bootcam

MongoDB Project

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Through this project, we've created a dashboard utilizing MongoDB using volcanos around the world dataset, where we did create multiple pipelines and charts to derive insights, initiating an overview over the volcanos around the world, and here are our main findings:

This dataset talks about Volcanoes on Earth and **includes 2,662 records**. We can tentatively conclude that we have: **20 countries** that have had volcanoes and, there **are 2,662 erupted volcanoes** the dataset which also describes the name, type, and image of the volcano, the country and region in which it occurred, its latitude and longitude and the population of the people affected by it.

Main insights:

- **Rakkibetsudake [Demon] volcano** in Japan has both the **highest mean latitude and longitude**, while **Kobi-sho volcano** has the **lowest longitude** and **Nikko volcano** with the **lowest mean latitude**.
- **Starto volcano type** had the **highest density of populations** around it, among all types within 5km radiant.
- **Starto volcano type** had the **highest occurrences among all volcano types with 31 occurrences**, where population within 10 km more than 100,000 people.
- The **United States** had the **highest Volcanoes erupted of all time with 323 eruptions**. However, **after the common era (i.e., after BCE)** **Russia eruptions rose with 214 eruptions**.
- **Submarine volcano type** has the **lowest mean elevation** with -5,700, which means that submarine happened under the sea levels, while **Starto volcano** has the **highest mean elevation**

Dashboard link:

<https://charts.mongodb.com/charts-project-0-metjc/dashboards/baa43de5-8782-4b14-92d2-6d4ff641bcf9>